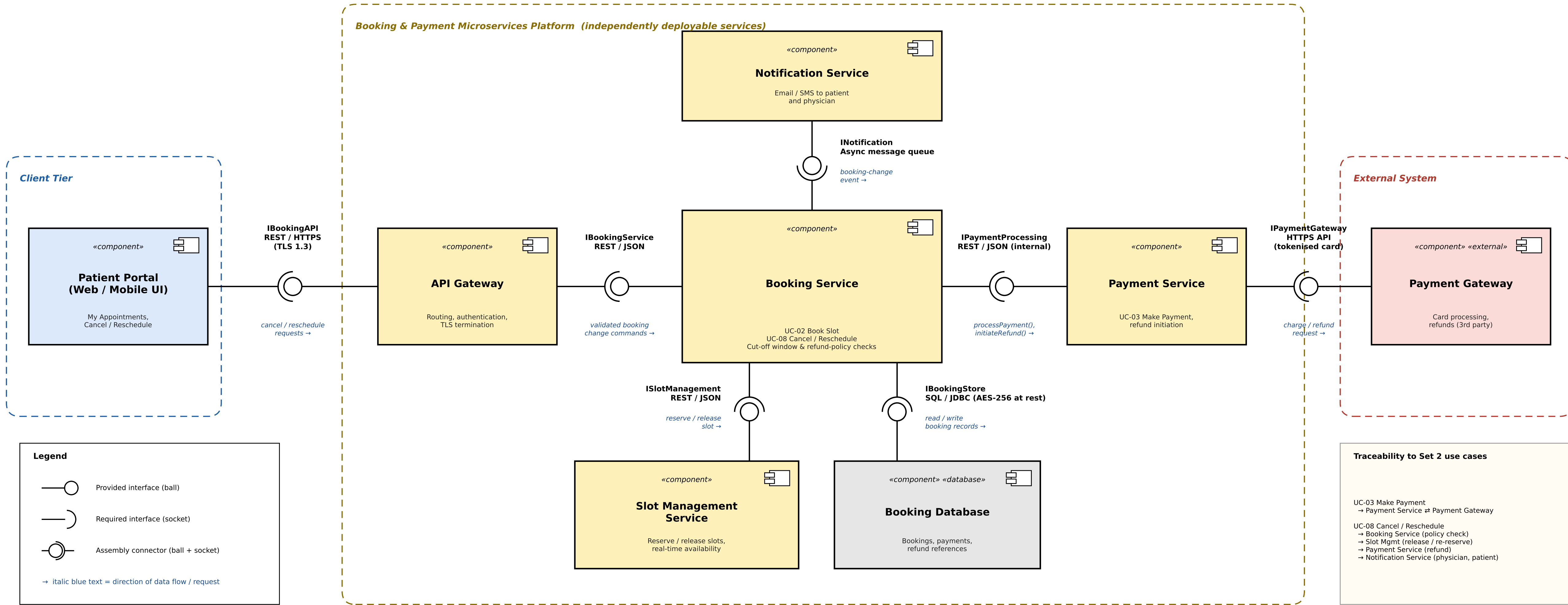


Telemedicine Slot Booking & Prescription Portal — UML Component Diagram

Microservices Architecture | Set 2: Payment & Booking Changes (UC-03 Make Payment, UC-08 Cancel / Reschedule Slot)



Lab 3: Architecture Selection & Justification

Name: Harsha R | SRN: PES1UG24AM115 | Telemedicine Slot Booking & Prescription Portal (Set 2: Payment & Booking Changes)

We chose Microservices Architecture for the Telemedicine Slot Booking & Prescription Portal.

Architectural Choice

The system is built as small, independently deployable services behind an API Gateway: Patient Portal (client), API Gateway, Booking Service, Payment Service, Slot Management Service, Notification Service and Booking Database, with the external Payment Gateway reached only through the Payment Service. Layered was rejected because one tightly coupled stack cannot scale video and booking separately; Client-Server was rejected because its single central server is a single point of failure.

Style	Main strength for this scenario	Main weakness for this scenario
Layered	Simple, clear separation of concerns	Hard to scale one part (e.g. video); layers are tightly coupled
Client-Server	Central control, easy data consistency	Single point of failure; risks the 99.9% uptime target
Microservices	Independent scaling and fault isolation	More operational complexity (accepted trade-off)

Two Reasons for the Choice

- **Independent scaling (NFR-002).** The portal must support 500 concurrent video sessions under 300 ms latency, while booking, payment and cancellation traffic follows a very different pattern. Separate services let each be scaled on its own, so a peak of Cancel / Reschedule requests (UC-08) never competes with video resources.
- **Fault isolation and no lost bookings (NFR-003).** If the external Payment Gateway or the Notification Service goes down, only that service is affected. Confirmed bookings stay safe in the Booking Service and Database, and a failed refund or notification can be retried later, supporting 99.9% uptime and zero lost records during failover.

Security Advantage

Payment isolation. Only the Payment Service talks to the external Payment Gateway, using tokenised card data, so card details are never stored or handled by the Booking Service or Database. This shrinks the attack surface and the compliance scope. The API Gateway is the single entry point enforcing authentication and TLS 1.3 (NFR-001), and the Booking Database encrypts patient and booking data at rest with AES-256.

Performance Benefit

Asynchronous, non-blocking interactions. The Booking Service publishes cancel / reschedule events to the Notification Service through a message queue, so the patient receives the booking-change confirmation immediately instead of waiting for email or SMS delivery to the physician. Each service can also be cached and scaled horizontally on its own, keeping response times low at peak clinic hours.